

VOI, Kenya

WMO: 637930

Lat: 3.398S

Lon: 38.558E

Elev: 1837

StdP: 13.75

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	64.6	65.8	52.3	62.3	81.2	54.0	66.3	80.9	30.9	77.0	28.0	78.2	10.1	270	0.387

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	19.6	94.2	72.0	92.8	71.9	91.2	71.9	76.1	85.5	75.4	84.7	74.8	83.8	11.6	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.8	135.1	78.3	73.4	132.9	77.9	72.7	130.0	77.5	40.9	85.2	40.3	85.1	39.7	84.3	86.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.0	24.3	21.8	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	57.5	78.8	2.6	2.7	55.6	80.8	54.0	82.4	52.6	83.9	50.7	85.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	78.3	79.9	81.0	82.1	80.7	77.8	76.2	74.6	74.4	75.8	78.2	79.3	79.3
	DBStd	3.16	1.82	2.19	2.01	2.26	2.10	1.85	1.67	1.90	1.96	2.23	2.15	1.81
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	10314	926	869	994	921	862	787	763	756	774	875	879	908
	CDD65	4839	461	449	529	471	397	337	298	291	324	410	429	443
	CDH74	36811	3912	4180	5064	3963	2670	2025	1603	1536	2113	3030	3299	3416
	CDH80	13408	1597	1920	2357	1558	788	499	294	286	603	1127	1214	1165
Wind	WSAvg	10.1	7.4	7.2	8.1	9.0	12.4	12.5	13.9	13.5	12.2	10.1	7.7	6.5
Precipitation	PrecAvg	22.7	1.4	0.9	3.6	3.5	1.1	0.2	0.1	0.2	0.6	1.2	5.2	4.6
	PrecMax	37.0	5.7	4.1	15.2	10.8	6.3	1.3	0.6	1.3	3.7	4.8	13.1	10.2
	PrecMin	11.0	0.0	0.0	0.3	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.7
	PrecStd	7.0	1.7	0.9	3.4	2.3	1.2	0.3	0.2	0.3	0.9	1.2	3.1	2.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.8	96.2	96.3	94.8	90.4	88.1	86.8	87.9	89.9	91.8	92.8	90.9
		MCWB	71.0	71.6	72.9	72.9	71.3	68.2	67.1	68.0	68.4	70.5	72.5	72.6
	2%	DB	91.5	93.9	94.3	92.1	88.0	86.3	84.9	84.8	87.6	89.9	90.2	89.2
		MCWB	71.2	71.4	72.3	73.2	70.9	67.5	66.7	67.3	67.9	70.2	71.9	72.5
	5%	DB	89.7	92.1	92.7	90.0	86.3	84.6	83.3	82.9	85.7	88.1	88.4	87.6
		MCWB	71.4	71.3	72.4	73.3	70.6	67.4	66.3	66.8	67.7	69.9	72.0	72.6
	10%	DB	87.8	90.0	90.9	88.0	84.6	83.0	81.5	81.3	83.8	86.2	86.5	86.1
		MCWB	71.7	71.3	72.6	73.2	70.3	67.4	66.1	66.6	67.4	69.6	72.0	72.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	75.9	76.5	77.1	76.7	74.6	71.5	70.2	70.9	73.1	74.7	75.6	76.5
		MCDB	85.2	85.4	88.4	86.2	83.7	80.6	81.6	78.7	83.0	82.6	83.0	84.2
	2%	WB	74.7	75.1	75.9	75.5	73.5	70.4	68.9	69.6	70.9	73.3	74.9	75.6
		MCDB	83.9	84.8	86.8	85.4	82.2	79.4	79.4	78.7	80.7	81.8	82.5	83.2
	5%	WB	73.9	74.1	75.2	74.7	72.6	69.7	68.0	68.7	69.8	72.4	74.3	74.8
		MCDB	83.1	83.9	85.7	84.3	81.1	78.6	78.3	78.2	79.6	81.1	81.8	82.3
	10%	WB	73.2	73.3	74.5	74.2	71.8	69.1	67.3	67.9	68.7	71.5	73.6	74.2
		MCDB	82.3	83.2	84.3	83.3	80.0	77.9	77.1	77.6	78.6	80.4	81.1	81.5
Mean Daily Temperature Range	5% DB	MDBR	17.8	19.5	19.6	16.8	14.9	15.7	16.0	15.6	17.4	17.8	16.5	15.3
		MCDBR	20.4	22.6	21.8	19.1	17.3	17.9	18.2	18.2	20.2	20.6	19.1	17.0
		MCWBR	4.9	5.0	4.6	4.1	4.3	4.3	4.5	4.5	5.1	4.8	4.3	4.3
	5% WB	MCDBR	17.3	18.8	18.9	16.9	14.3	15.2	15.9	15.6	17.1	17.3	15.8	15.3
		MCWBR	5.0	5.3	4.9	4.6	4.8	4.4	4.7	4.8	5.9	5.4	4.9	4.9
Clear-Sky Solar Irradiance	taub		0.437	0.453	0.420	0.369	0.364	0.395	0.405	0.391	0.381	0.382	0.390	0.424
	taud		2.312	2.254	2.377	2.560	2.558	2.426	2.362	2.391	2.416	2.436	2.452	2.375
	Ebn at Noon		285	281	287	295	288	274	273	284	294	298	297	288
	Edh at Noon		43	46	41	33	31	35	38	38	38	38	37	40
All-Sky Solar Radiation	RadAvg		1940	2002	1956	1743	1545	1446	1441	1539	1766	1886	1861	1885
	RadStd		101	92	114	87	77	60	87	85	88	89	97	80

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	+0.65	+0.85	+0.83	+0.56	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page